

Pin connection diagram for ESP32-53-WROOM-1U to NI682. The diagram shows a breadboard with two rows of pins. The top row (pins 1-16) is connected to the ESP32: EN (1), TXD0 (2), RXD0 (3), RXD0 (4), TXD0 (5), TXD0 (6), TXD0 (7), TXD0 (8), TXD0 (9), TXD0 (10), TXD0 (11), TXD0 (12), TXD0 (13), TXD0 (14), TXD0 (15), TXD0 (16). The bottom row (pins 17-32) is connected to the NI682: EN (17), TXD0 (18), RXD0 (19), RXD0 (20), TXD0 (21), TXD0 (22), TXD0 (23), TXD0 (24), TXD0 (25), TXD0 (26), TXD0 (27), TXD0 (28), TXD0 (29), TXD0 (30), TXD0 (31), TXD0 (32). The diagram also shows the ESP32's TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0 pins connected to the NI682's TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0, TXD0, RXD0 pins.

The diagram shows the pin connections for the LCD module:

- J5 DISPLAY**:
 - Pin 1: +5V
 - Pins 2-8: GND
 - Pin 9: LCD_CS_B
 - Pin 10: LCD_RST
 - Pin 11: LCD_MOSI_R
 - Pin 12: LCD_SCK
 - Pin 13: LCD_SDA
 - Pin 14: LCD_SCL
- Buttons**:
 - BTN_UP: Pin 16 NAV_SW 1
 - BTN_DOWN: Pin 16 NAV_SW 2
 - BTN_LEFT: Pin 16 NAV_SW 3
 - BTN_RIGHT: Pin 16 NAV_SW 4
 - BTN_SEL: Pin 16 NAV_SW 5
 - Pin 16 NAV_SW 6: GND
- UART**:
 - +3V3: TXD0 (Pin 17 AUX_2)
 - GND: RXD0 (Pin 17 AUX_1)
 - +3V3: TXD1 (Pin 17 AUX_4)
 - GND: RXD1 (Pin 17 AUX_3)
 - +3V3: I2C_SCL (Pin 17 AUX_6)
 - GND: I2C_SDA (Pin 17 AUX_5)

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SSR terminals 14/9J: $j_{p1} = SSR_{EN}(\text{board} + \text{v5v, watchdog-gated})$
 $j_{p2} = \text{the switched low side, Board-safe: } H720/\text{hold each} \\
\text{input gate down while the E5372 pins are high impedance.}$
 NOT optional: $j_{p3} = \text{level 1 and level 2 generated: it an opto}$
 $\text{only isolates if the SSR loop is powered off-and-on: an opto}$
 both powers it.

TC1 terminal J3 / TC2 terminal J8: $j_{p1} = K_{+}$, $j_{p2} = K_{-}$ – both
 tied, biased near AGND only through each MAX485's internal
 BIOS terminal. Ungrounded-junction boards required: two
 grounded-junction boards in one kin would loop through it.

Nav switch J6 is panel-mounted; Inputs use ESP32 pull-ups.

J11: $IN1/IN2 \rightarrow IN3$ (lilo/gas-flow/spare) + GND – dry contact each
 channel to GND.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted – do not wire to AC mains. Y1/C25/C26 load caps are an ASSUM value (unverified C_L). see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.

4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR +
Bisque project
Sheet: /
File: bisque-controller.kicad_sch
Title: Bisque Kiln Controller
Size: A1 Date: 2026-07-20
KiCad E.D.A. 10.0.6